Class #1 - Eigenvectors and Eigenvalues

1. Example: Consider the system of DE

$$\frac{dx(t)}{dt} = Ax(t).$$

Solution to above depends on the eigenvalues of the matrix A.

2. Def: Given a matrix $A \in$ $\mathbb{R}^{n \times n}$, a nonzero vector x such that $\boxed{Ax = \lambda x}$ for some scalar λ is called an eigenvector of A.

vector, scalar (real or complex number), vector

will hold if $x=0$ for any A and any λ

And the corresponding scalar λ is called eigenvalue.

3. Example: Let A be an $n \times n$ zero matrix. Find an eigenvector. Is $\lambda = 1$ an eigenvalue of A?

$Ax = \lambda x$, $x \neq 0$

$\lambda \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \lambda x_1 \\ \vdots \\ \lambda x_n \end{bmatrix}$

If $\lambda = 0$, then every nonzero vector x is an eigenvector of A.

$Ax = 0 \cdot x$

No! $Ax = 0$

$\lambda x = 1 \cdot x = x \neq 0$ (to be an eigenvector, it must be nonzero)

$T(x) = Ax$

4. A scalar λ is called an eigenvalue of A if there is a nontrivial solution to the equation

$$Ax = \lambda x.$$

$\Longleftrightarrow \quad (A - \lambda I)x = 0.$

$Ax - \lambda x = 0$

$(A - \lambda I)x = Ax - \lambda x$

That nontrivial solution (nonzero) is called an eigenvector of A corresponding to the eigenvalue of A.

5. Example: Is $\begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix}$ an eigenvector of $\begin{bmatrix} 3 & 7 & 9 \\ -4 & -5 & 1 \\ 2 & 4 & 4 \end{bmatrix}$? If so, find the eigenvalue.

Check if $Ax = \lambda x$ for some scalar λ.

$\begin{bmatrix} 3 & 7 & 9 \\ -4 & -5 & 1 \\ 2 & 4 & 4 \end{bmatrix}\begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix} \stackrel{?}{=} \lambda \begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix}$

$= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

If $\lambda = 0$, this is true.

Ans: Yes.

eigenvalue $= 0$.

6. Example: Is $\lambda = 4$ an eigenvalue of $\begin{bmatrix} 3 & 0 & -1 \\ 2 & 3 & 1 \\ -3 & 4 & 5 \end{bmatrix}$? If so, find one corresponding eigenvector.

$x = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$

Yes because $Ax = 4x$ has a nontrivial soln.

Qn: $Ax = 4x$ has a nontrivial soln?

$\iff (A - 4I)x = 0$ has a nontrivial soln?

$$A - 4I = \begin{bmatrix} 3-4 & 0 & -1 \\ 2 & 3-4 & 1 \\ -3 & 4 & 5-4 \end{bmatrix} = \begin{bmatrix} -1 & 0 & -1 \\ 2 & -1 & 1 \\ -3 & 4 & 1 \end{bmatrix} = B$$

$\text{Nul } B = \{ x \in \mathbb{R}^3 \mid Bx = 0 \} = \{ (x_1, x_2, x_3) \mid x_1 = -t,\ x_2 = -t,\ x_3 = t,\ t \in \mathbb{R} \}$

Have to solve $Bx = 0$. $\boxed{Bx=0}$

$$\begin{bmatrix} \boxed{-1} & 0 & -1 \\ 2 & -1 & 1 \\ -3 & 4 & 1 \end{bmatrix} \underset{\substack{r_2 \leftarrow r_2 + 2 \cdot r_1 \\ r_3 \leftarrow r_3 + (-3) r_1}}{\sim} \begin{bmatrix} -1 & 0 & -1 \\ 0 & -1 & -1 \\ 0 & 4 & 4 \end{bmatrix} \sim \begin{array}{ccc} x_1 & x_2 & x_3 \\ \end{array} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

nonpivot column.

To have a nontrivial soln, need a nonpivot col.

$\therefore$ Columns of $A - \lambda I$ must be L. D. to have an eigenvector corresponding to λ

7. Note: λ is an eigenvalue of if the equation $\underline{Ax = \lambda x}$, i.e. $\underline{(A - \lambda I)x = 0}$ has a __nontrivial__ solution. Hence, λ is an eigenvalue of A if and only if columns of the matrix $\underline{A - \lambda I}$ are __L. D.__, i.e., $\underline{\det(A - \lambda I)} = 0$.

① See if there exists λ such that $\det(A - \lambda I) = 0$.

② If so, put that λ into the equation $Ax = \lambda x$ or $(A - \lambda I)x = 0$, and $\boxed{\text{find a nontrivial soln.}}$

A in $\mathbb{R}^{n \times n}$

$A - \lambda I$ in $\mathbb{R}^{n \times n}$

8. Def: The set of all solutions to the equation

// $\text{Nul}(A - \lambda I)$ $\qquad (A - \lambda I)x = \underline{\ 0\ }$

is called $\boxed{\textit{eigenspace of } A \textit{ corresponding to } \lambda.}$

$B \in \mathbb{R}^{n \times m}$

$\text{Nul } B = \{ x \in \mathbb{R}^m \mid Bx = 0 \}$ is a subspace of $\mathbb{R}^m$

• Subset of $\mathbb{R}^m$
• Contains zero vector
• closed under addition
• closed under scalar multiplication

$B0 = 0$

$B(u+v) = Bu + Bv = 0$, $u, v \in \text{Nul } B$

$B(cu) = cB(u) = 0$, $u \in \text{Nul } B$

$\text{Range } T = \text{Col } B = \{ b \in \mathbb{R}^n \mid Bx = b \}$ is a subspace of $\mathbb{R}^n$

$(T(x) = Bx)$

9. Note: eigenspace of A corresponding to λ = __Nul__$(A - \lambda I)$ which is a subspace of $\underline{\mathbb{R}^n}$.

$\dim \text{Nul}(A - \lambda I)$

10. Note: dim eigenspace of A corresponding to λ = nulllity of $\underline{A - \lambda I}$, and this number is called the __geometric__ *multiplicity of* λ.

11. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}$. Without any calculation, show that $\lambda = 0$ is an eigenvalue of A. Find a basis for the eigenspace corresponding to $\lambda = 0$. What is the geometric multiplicity of $\lambda = 0$? 2

$\det A = \boxed{0}$ because columns of A are L.D. $\therefore$ $A\mathbf{x} = \mathbf{0}$ has a nontrivial soln. $\therefore$ Nul A contains a nonzero vector. $\therefore$ $\lambda = \boxed{0}$ is an eigenvalue of A.

Nul(A) $=$ Nul$(A - 0 \cdot I)$

$A\mathbf{x} = \lambda\mathbf{x} \iff (A - \lambda I)\mathbf{x} = \mathbf{0}$ has a nontrivial soln. $\iff \det(A - \lambda I) = 0$

$\det A = \det(A - 0 \cdot I) = 0$

$\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$

geometric multiplicity $= \dim \text{Nul}(A - 0 \cdot I)$

Fill in the blank: Any nonzero multiple of an eigenvector is also an eigenvector corresponding to the same eigenvalue.

$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}$ $\quad \det A = 0 = \det(A - 0 \cdot I)$ (A has L.D. columns)

$\therefore \lambda = 0$ is an eigenvalue of A

Eigenspace of A corresponding to $\lambda = 0$: Nul$(A - 0 \cdot I)$

$\dim \text{Nul}(A - 0 \cdot I) = \dim \text{Nul} A = \text{nullity } A$
$= \#$ nonpivot col.
$= 2$

How many vectors are in any basis of Eigenspace of A corresponding to $\lambda = 0$?

Have: $A\mathbf{x} = \lambda\mathbf{x}$ ($\mathbf{x} \neq 0$ eigenvector, λ scalar)

To show: $A(c\mathbf{x}) = \lambda(c\mathbf{x})$, $c \neq 0$

Since $\mathbf{x} \neq 0$, $c\mathbf{x} \neq 0$ for any nonzero scalar c.
Let $\mathbf{y} = c\mathbf{x}$. Then $\mathbf{y} \neq 0$.
and $c(A\mathbf{x}) = c(\lambda\mathbf{x})$
$\iff A(c\mathbf{x}) = \lambda(c\mathbf{x})$
$\iff A\mathbf{y} = \lambda\mathbf{y}$
$\therefore$ $\mathbf{y}$ is also an eigenvector corresponding to the same eigenvalue λ.

12. Eigenvalues of a triangular matrix:

$n \times n$ $\quad \det A = \det A^T$

Eigenvalues are when $\det(A - \lambda I) = 0$.

$A = \begin{bmatrix} a_{11} & & 0 \\ & a_{22} & \\ * & & a_{nn} \end{bmatrix}$ $\quad \begin{bmatrix} a_{11} & & 0 \\ & a_{22} & \\ * & & a_{nn} \end{bmatrix}$ Lower triangular

$\begin{bmatrix} a_{11} & & * \\ & \ddots & \\ 0 & & a_{nn} \end{bmatrix}$ upper-triangular

$\Rightarrow A - \lambda I = \begin{bmatrix} a_{11} - \lambda & & 0 \\ & a_{22} - \lambda & \\ * & & a_{nn} - \lambda \end{bmatrix}$ $\quad \det(A - \lambda I) = 0$

$(5-\lambda)(2-\lambda)(4-\lambda) = 0$

$\det(A - \lambda I) = \prod_{i=1}^{n} (a_{ii} - \lambda) = 0$

$\iff \lambda = a_{ii} \; \forall i$

Recall: det of a triangular matrix is prod. of diagonal elements.

Fill in the blank: If A is an $n \times n$ triangular matrix, then a diagonal element is an eigenvalue of A.

ex $\quad I_{3 \times 3}$

eigenvalue eigenvector

nonzero.

scalar (real or complex numbers)

A $n \times n$

$$Ax = \lambda x$$

$$\Longleftrightarrow (A - \lambda I)x = 0$$

$(A - \lambda I)x = 0$ has a nontrivial soln

$$\Longleftrightarrow \det(A - \lambda I) = 0$$

If λ is an eigenvalue,

$\mathrm{Nul}(A - \lambda I)$

is called

eigenspace of A corresponding to λ.

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① Attendance

② Par. quiz

③ Class #1 handout #10 ~

④ **Quiz #1** Next Mon 9/14 3:00 – 3:10 pm PP #1 ~ #3

① $A = \begin{bmatrix} 3 & 0 & -1 \\ 2 & 3 & 1 \\ -3 & 4 & 5 \end{bmatrix}$

$$\mathrm{Nul}(A - 4I) = \left\{ x \in \mathbb{R}^3 \;\middle|\; x = \begin{bmatrix} -t \\ -t \\ t \end{bmatrix} \right\}$$

$$(A - 4I)\begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} = 0 \qquad = \left\{ t \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \;\middle|\; t \in \mathbb{R} \right\}$$

$$\Rightarrow \quad A \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} = 4 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

T/F ~~T~~ $\begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$ is an eigenvector of A corresponding to eigenvalue $\lambda = \boxed{4}$.

$A \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = 4 \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow$ T/F F $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ is an e-vector corresponding to e-value $\lambda = 4$.

$Nul(A - 4I) = \left\{ t \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \;\middle|\; t \in \mathbb{R} \right\} = \text{span}\left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \right\}$

T/F A spanning set of $Nul(A-4I)$ is $\left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}$ True

① L.I ② spans the subspace

T/F A basis of $Nul(A-4I)$ is $\left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \right\}$. T

$\left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}$

since a subspace

T/F False Nullity of $(A-4I)$ is $\cancel{2}$. 1

T/F $Nul(A-4I)$ is called an eigenspace of A corresponding to $\lambda = 4$. True

T/F Every non vector in eigenspace is an eigenvector. False